

Section	Line	Quantity	Value (mean ± SEM)	n	Source
A. Canonical recovery (Fig 3A)	218	MSE_pB @ noise=0, pooled, seed (baseline)	0.0658 ± 0.0059	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery (Fig 3A)	218	MSE_pB @ noise=0, pooled, surfaced	0.0127 ± 0.0044	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery (Fig 3A)	218	MSE_pB @ noise=0, pooled, best-surfaced	0.0025 ± 0.0011	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery (Fig 3A)	218	MSE_pB @ noise=0, pooled, ceiling (GT vs itself)	0.0006 ± 0.0000	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery (Fig 3A)	218	MSE_pB @ noise=0, pooled, floor (random policy)	0.0894 ± 0.0066	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery vs noise (Fig 3A)	223	MSE_pB @ noise=0.0, pooled, surfaced	0.0127 ± 0.0044	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery vs noise (Fig 3A)	223	MSE_pB @ noise=0.0, pooled, seed	0.0658 ± 0.0059	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery vs noise (Fig 3A)	223	MSE_pB @ noise=0.0, pooled, best-surfaced	0.0025 ± 0.0011	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery vs noise (Fig 3A)	223	MSE_pB @ noise=0.5, pooled, surfaced	0.0113 ± 0.0011	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery vs noise (Fig 3A)	223	MSE_pB @ noise=0.5, pooled, seed	0.0667 ± 0.0061	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery vs noise (Fig 3A)	223	MSE_pB @ noise=0.5, pooled, best-surfaced	0.0062 ± 0.0010	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery vs noise (Fig 3A)	223	MSE_pB @ noise=0.75, pooled, surfaced	0.0289 ± 0.0042	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery vs noise (Fig 3A)	223	MSE_pB @ noise=0.75, pooled, seed	0.0657 ± 0.0060	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. Canonical recovery vs noise (Fig 3A)	223	MSE_pB @ noise=0.75, pooled, best-surfaced	0.0221 ± 0.0045	15.0	recovery_vs_clean_gt/recovery_long.csv (MSE_pB vs c
A. LLM-judge mechanism similarity (Fig 3B)	226	judge similarity (0-1) @ noise=0.0, pooled	0.7817 ± 0.0564	30.0	*_sampling/noise=*/judge_similarity_desc.csv (descr
A. LLM-judge mechanism similarity (Fig 3B)	226	judge similarity (0-1) @ noise=0.5, pooled	0.6589 ± 0.0457	30.0	*_sampling/noise=*/judge_similarity_desc.csv (descr
A. LLM-judge mechanism similarity (Fig 3B)	226	judge similarity (0-1) @ noise=0.75, pooled	0.6011 ± 0.0573	30.0	*_sampling/noise=*/judge_similarity_desc.csv (descr
A. LLM-judge mechanism similarity (Fig 3B)	226	judge similarity (0-1) @ noise=1.0, pooled	0.2566 ± 0.0591	30.0	*_sampling/noise=*/judge_similarity_desc.csv (descr
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB alternating / seed	0.1071 ± 0.0014	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB alternating / surfaced	0.0042 ± 0.0008	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB alternating / best-surfaced	0.0038 ± 0.0005	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB alternating / ceiling	0.0039 ± 0.0005	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB alternating / floor	0.0033 ± 0.0000	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB perseveration / seed	0.1208 ± 0.0045	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB perseveration / surfaced	0.0032 ± 0.0005	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB perseveration / best-surfaced	0.0024 ± 0.0003	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB perseveration / ceiling	0.0049 ± 0.0012	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB perseveration / floor	0.0017 ± 0.0000	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB take_the_worst / seed	0.1481 ± 0.0048	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB take_the_worst / surfaced	0.0671 ± 0.0332	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB take_the_worst / best-surfaced	0.0191 ± 0.0145	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB take_the_worst / ceiling	0.0020 ± 0.0002	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB take_the_worst / floor	0.1266 ± 0.0000	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB single_cue / seed	0.1748 ± 0.0018	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB single_cue / surfaced	0.1321 ± 0.0510	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB single_cue / best-surfaced	0.1211 ± 0.0510	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB single_cue / ceiling	0.0038 ± 0.0007	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB single_cue / floor	0.0656 ± 0.0000	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB anti_majority / seed	0.3612 ± 0.0063	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB anti_majority / surfaced	0.2142 ± 0.0689	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB anti_majority / best-surfaced	0.1644 ± 0.0671	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB anti_majority / ceiling	0.0025 ± 0.0001	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB anti_majority / floor	0.0943 ± 0.0000	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB cue_parity / seed	0.1929 ± 0.0022	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB cue_parity / surfaced	0.1721 ± 0.0124	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB cue_parity / best-surfaced	0.1517 ± 0.0138	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB cue_parity / ceiling	0.0026 ± 0.0002	3.0	per_model/recovery_long.csv (figure source: MSE_pB
B. Non-canonical recovery (Fig 3C)	231-232	MSE_pB cue_parity / floor	0.1320 ± 0.0000	3.0	per_model/recovery_long.csv (figure source: MSE_pB
C. Case study 1 survivor pi_4 NLSWM (Fig 4C/D)	273	mse_pB pi_4 @ cycle 5 (final)	0.0097 ± 0.0018	10.0	ttb+wadd/analysis/eval_run_self/trajectory_all.csv
C. Case study 1 survivor pi_4 NLSWM (Fig 4C/D)	273	pearson_r_pB pi_4 @ cycle 5 (final)	0.9168 ± 0.0343	10.0	ttb+wadd/analysis/eval_run_self/trajectory_all.csv
C. Case study 1 best-surfaced trajectory (Fig 4C/D)	273	best surfaced mse_pB cycle 1 -> 5	0.0933 -> 0.0097	nan	ttb+wadd/analysis/eval_run_self/trajectory_all.csv
C. Case study 1 best-surfaced trajectory (Fig 4C/D)	273	best surfaced pearson_r_pB cycle 1 -> 5	0.3546 -> 0.9168	nan	ttb+wadd/analysis/eval_run_self/trajectory_all.csv
C. Case study 1 per-cycle best-surfaced	273	mse_pB best-surfaced @ cycle 1 (pi_3)	0.0933 ± 0.0330	10.0	ttb+wadd/analysis/eval_run_self/per_experiment_summa
C. Case study 1 per-cycle best-surfaced	273	mse_pB best-surfaced @ cycle 2 (pi_4)	0.0097 ± 0.0018	10.0	ttb+wadd/analysis/eval_run_self/per_experiment_summa
C. Case study 1 per-cycle best-surfaced	273	mse_pB best-surfaced @ cycle 3 (pi_4)	0.0097 ± 0.0018	10.0	ttb+wadd/analysis/eval_run_self/per_experiment_summa
C. Case study 1 per-cycle best-surfaced	273	mse_pB best-surfaced @ cycle 4 (pi_4)	0.0097 ± 0.0018	10.0	ttb+wadd/analysis/eval_run_self/per_experiment_summa
C. Case study 1 per-cycle best-surfaced	273	mse_pB best-surfaced @ cycle 5 (pi_4)	0.0097 ± 0.0018	10.0	ttb+wadd/analysis/eval_run_self/per_experiment_summa
C. Case study 1 per-cycle best-surfaced	273	pearson_r_pB best-surfaced @ cycle 1 (pi_3)	0.3546 ± 0.1377	10.0	ttb+wadd/analysis/eval_run_self/per_experiment_summa
C. Case study 1 per-cycle best-surfaced	273	pearson_r_pB best-surfaced @ cycle 2 (pi_4)	0.9168 ± 0.0343	10.0	ttb+wadd/analysis/eval_run_self/per_experiment_summa
C. Case study 1 per-cycle best-surfaced	273	pearson_r_pB best-surfaced @ cycle 3 (pi_4)	0.9168 ± 0.0343	10.0	ttb+wadd/analysis/eval_run_self/per_experiment_summa
C. Case study 1 per-cycle best-surfaced	273	pearson_r_pB best-surfaced @ cycle 4 (pi_4)	0.9168 ± 0.0343	10.0	ttb+wadd/analysis/eval_run_self/per_experiment_summa
C. Case study 1 per-cycle best-surfaced	273	pearson_r_pB best-surfaced @ cycle 5 (pi_4)	0.9168 ± 0.0343	10.0	ttb+wadd/analysis/eval_run_self/per_experiment_summa
C. Case study 1 cycle effect (LME)	273	mse_pB ~ cycle + (1 experiment): beta_cycle	-0.01391 (SE 0.00430), z=-3.24, p=0.0012, n=90	10.0	ttb+wadd/analysis/eval_run_self/per_experiment_summa
C. Case study 1 cycle effect (LME)	273	pearson_r_pB ~ cycle + (1 experiment): beta_cycle	+0.08587 (SE 0.02652), z=+3.24, p=0.00121, n=90	10.0	ttb+wadd/analysis/eval_run_self/per_experiment_summa
C. Case study 1 final leaderboard (Fig 4j)	273	pi_1 (base): MSE / r	MSE=0.0760, r=0.780, MAE=0.202	24.0	ttb+wadd/analysis/eval_hilbig/eval_hilbig_summary.cs
C. Case study 1 final leaderboard (Fig 4j)	273	pi_2 (base): MSE / r	MSE=0.0537, r=0.913, MAE=0.204	24.0	ttb+wadd/analysis/eval_hilbig/eval_hilbig_summary.cs
C. Case study 1 final leaderboard (Fig 4j)	273	pi_4 (surfaced): MSE / r	MSE=0.0377, r=0.958, MAE=0.187	24.0	ttb+wadd/analysis/eval_hilbig/eval_hilbig_summary.cs
C. Case study 1 final leaderboard (Fig 4j)	273	pi_7 (surfaced): MSE / r	MSE=0.0278, r=0.926, MAE=0.099	24.0	ttb+wadd/analysis/eval_hilbig/eval_hilbig_summary.cs
C. Case study 1 reliability floor (Fig 4D)	273	human-noise MSE floor	0.0033	70.0	calibration_points_all.csv: mean(p_b_human_sem^2)
D. Case study 2 per-cycle best-surfaced	354	mse_pB best-surfaced @ cycle 1 (pi_3)	0.0371 ± 0.0069	10.0	ttb+tallying/analysis/eval_run_self/per_experiment_s
D. Case study 2 per-cycle best-surfaced	354	mse_pB best-surfaced @ cycle 2 (pi_3)	0.0371 ± 0.0069	10.0	ttb+tallying/analysis/eval_run_self/per_experiment_s
D. Case study 2 per-cycle best-surfaced	354	mse_pB best-surfaced @ cycle 3 (pi_5)	0.0210 ± 0.0063	10.0	ttb+tallying/analysis/eval_run_self/per_experiment_s
D. Case study 2 per-cycle best-surfaced	354	mse_pB best-surfaced @ cycle 4 (pi_6)	0.0188 ± 0.0033	10.0	ttb+tallying/analysis/eval_run_self/per_experiment_s
D. Case study 2 per-cycle best-surfaced	354	mse_pB best-surfaced @ cycle 5 (pi_7)	0.0180 ± 0.0051	10.0	ttb+tallying/analysis/eval_run_self/per_experiment_s
D. Case study 2 per-cycle best-surfaced	354	pearson_r_pB best-surfaced @ cycle 1 (pi_3)	0.7767 ± 0.0594	10.0	ttb+tallying/analysis/eval_run_self/per_experiment_s
D. Case study 2 per-cycle best-surfaced	354	pearson_r_pB best-surfaced @ cycle 2 (pi_3)	0.7767 ± 0.0594	10.0	ttb+tallying/analysis/eval_run_self/per_experiment_s
D. Case study 2 per-cycle best-surfaced	354	pearson_r_pB best-surfaced @ cycle 3 (pi_3)	0.7767 ± 0.0594	10.0	ttb+tallying/analysis/eval_run_self/per_experiment_s
D. Case study 2 per-cycle best-surfaced	354	pearson_r_pB best-surfaced @ cycle 4 (pi_6)	0.8032 ± 0.0368	10.0	ttb+tallying/analysis/eval_run_self/per_experiment_s
D. Case study 2 per-cycle best-surfaced	354	pearson_r_pB best-surfaced @ cycle 5 (pi_6)	0.8032 ± 0.0368	10.0	ttb+tallying/analysis/eval_run_self/per_experiment_s
D. Case study 2 cycle effect (LME)	354	mse_pB ~ cycle + (1 experiment): beta_cycle	-0.00625 (SE 0.00140), z=-4.45, p=8.5e-06, n=90	10.0	ttb+tallying/analysis/eval_run_self/per_experiment_s
D. Case study 2 cycle effect (LME)	354	pearson_r_pB ~ cycle + (1 experiment): beta_cycle	+0.03161 (SE 0.01500), z=+2.11, p=0.0351, n=90	10.0	ttb+tallying/analysis/eval_run_self/per_experiment_s
D. Case study 2 improvement over cycles (Fig 5C/D)	354	mse_pB WADD(c1) -> DR-WADD(c5)	0.0371 -> 0.0180	nan	ttb+tallying/analysis/eval_run_self/trajectory_all.c
D. Case study 2 improvement over cycles (Fig 5C/D)	354	pearson_r_pB WADD(c1) -> DR-WADD(c5)	0.7767 -> 0.7666	nan	ttb+tallying/analysis/eval_run_self/trajectory_all.c
D. Case study 2 final survivors (Fig 5C/D)	354	mse_pB pi_6 Satisficing WADD @ final cycle	0.0188 ± 0.0033	10.0	ttb+tallying/analysis/eval_run_self/trajectory_all.c
D. Case study 2 final survivors (Fig 5C/D)	354	pearson_r_pB pi_6 Satisficing WADD @ final cycle	0.8032 ± 0.0368	10.0	ttb+tallying/analysis/eval_run_self/trajectory_all.c
D. Case study 2 final survivors (Fig 5C/D)	354	mse_pB pi_7 Diminishing>Returns WADD @ final cycle	0.0180 ± 0.0051	10.0	ttb+tallying/analysis/eval_run_self/trajectory_all.c
D. Case study 2 final survivors (Fig 5C/D)	354	pearson_r_pB pi_7 Diminishing>Returns WADD @ final c	0.7666 ± 0.0798	10.0	ttb+tallying/analysis/eval_run_self/trajectory_all.c
D. Case study 2 pooled scatter (Fig 5E/F)	372	pi_6 Satisficing WADD: pooled r	0.707	134.0	ttb+tallying/analysis/eval_run_self/calibration_poin
D. Case study 2 pooled scatter (Fig 5E/F)	372	pi_6 Satisficing WADD: pooled MSE / MAE	MSE=0.0195, MAE=0.114	134.0	ttb+tallying/analysis/eval_run_self/calibration_poin
D. Case study 2 ambivalence (Fig 5E/F)	372	pi_6 Satisficing WADD: predictions in p(B) in [0.45,	72/134 = 53.7%	134.0	ttb+tallying/analysis/eval_run_self/calibration_poin
D. Case study 2 pooled scatter (Fig 5E/F)	372	pi_7 Diminishing>Returns WADD: pooled r	0.685	134.0	ttb+tallying/analysis/eval_run_self/calibration_poin
D. Case study 2 pooled scatter (Fig 5E/F)	372	pi_7 Diminishing>Returns WADD: pooled MSE / MAE	MSE=0.0216, MAE=0.109	134.0	ttb+tallying/analysis/eval_run_self/calibration_poin
D. Case study 2 ambivalence (Fig 5E/F)	372	pi_7 Diminishing>Returns WADD: predictions in p(B) i	27/134 = 20.1%	134.0	ttb+tallying/analysis/eval_run_self/calibration_poin
E. Centaur held-out Hilbig (line 1140)	1140	MAE seed Tallying	0.2207	24.0	centaur_seeds_tallying_ttb/analysis/eval_hilbig/eval
E. Centaur held-out Hilbig (line 1140)	1140	MAE seed TTB	0.2022	24.0	centaur_seeds_tallying_ttb/analysis/eval_hilbig/eval
E. Centaur held-out Hilbig (line 1140)	1140	MAE surfaced (winner) Non-linear Compensatory Attent	0.1314	24.0	centaur_seeds_tallying_ttb/analysis/eval_hilbig/eval
E. Centaur held-out Hilbig (line 1140)	1140	MAE surfaced (other) Scale-Invariant Heuristic Selec	0.2176	24.0	centaur_seeds_tallying_ttb/analysis/eval_hilbig/eval
F. Preregistration H1 model discrimination (Fig 5)	493 / Sig.Stm	pi_7 match-rate vs WADD (vs 0.5)	0.632 ± 0.076, z=1.74, p=0.0824	nan	preregistration_visualization/data/h1_results.csv (m
F. Preregistration H1 model discrimination (Fig 5)	493 / Sig.Stm	pi_7 match-rate vs TALLYING (vs 0.5)	0.678 ± 0.063, z=2.83, p=0.00472	nan	preregistration_visualization/data/h1_results.csv (m
F. Preregistration H1 model discrimination (Fig 5)	493 / Sig.Stm	pi_7 match-rate vs TTB (vs 0.5)	0.634 ± 0.068, z=1.97, p=0.0488	nan	preregistration_visualization/data/h1_results.csv (m
F. Preregistration H2 steep-vs-flat (Fig 5)	493 / Sig.Stm	steep-region choice rate vs 0.5	mean=0.582 ± 0.020, t(99)=4.19, p=6.03e-05, d=0.42	100.0	h2_participants.csv: one-sample t-test
F. Preregistration H3 diminishing preference (Fig 5)	493 / Sig.Stm	level-shift effect vs 0	mean=0.0291 ± 0.0160, t(99)=1.82, p=0.0715, d=0.18; Wilk	100.0	h3_participants.csv: one-sample t-test + Wilcoxon